

OBJECT ORIENTED PROGRAMMING LAB
ASSIGNMENT-4
(Topic – Inheritance)

Programming Questions:

- 1. Create a C++ program to demonstrate Single Inheritance.**
Create a base class Person with the following data members:
Name
Age
Create a derived class Student having:
Roll Number
Course
Implement member functions to accept and display the complete details of a student using inheritance.

- 2. Create a C++ program to demonstrate Multilevel Inheritance.**
Create three classes:
Person → stores Name
Employee → stores Employee ID
Manager → stores Department
Derive Employee from Person and Manager from Employee. Display all the details of the manager.

- 3. Create a C++ program to demonstrate Multiple Inheritance.**
Create two base classes:
Academic → stores Marks
Sports → stores Sports Score
Create a derived class Result that inherits from both classes and calculates the total score and displays the result.

- 4. Create a C++ program to demonstrate Hierarchical Inheritance.**
Create a base class Vehicle containing:
Brand
Model
Create two derived classes:
Car → Number of Doors
Bike → Engine Capacity
Create separate objects of Car and Bike and display their respective details.

5. Create a **base class College** with a constructor that initializes the College Name. Create a **derived class Department** with a constructor that initializes the Department Name.

Create an object of the derived class and display both college and department information to show the order of constructor execution.